

DARS: A Dynamic Adaptive Rating System for Intelligent Assessment with Graph-Based Item Routing, Contextual Remediation, and Performance Prediction

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Abstract—Proficiency tracking in adaptive assessment systems has long relied on fixed- K Elo ratings that treat every answered item identically, ignoring the learner’s evolving uncertainty, response quality, streak behaviour, and the structural position of an item in the knowledge graph. We propose DARS (Dynamic Adaptive Rating System), a unified backend framework built around a novel rating model that draws inspiration from competitive matchmaking systems (Valorant RR, TrueSkill) while being grounded in psychometric theory. DARS introduces four interdependent innovations: (i) a *dynamic uncertainty-decaying update factor* $K(n, \sigma^2)$ that starts high during a learner’s provisional phase and contracts as ratings stabilise, analogous to rating deviation in Glicko-2; (ii) a *multi-signal performance quality score* $\mathcal{P}$ that combines binary correctness, time efficiency, hint usage, and streak state into a continuous performance signal beyond pass/fail; (iii) a *streak multiplier* ϕ that rewards sustained high performance and penalises consistent failure, mirroring competitive rank-rate bonuses; and (iv) a *knowledge-graph-aware difficulty calibration* that uses betweenness centrality to modulate update magnitude for hub items. These are integrated with a directed knowledge graph for adaptive item routing, a proximity-constrained remediation engine, an answer-velocity anomaly detector, and a four-component performance predictor. The complete system is formally specified and domain-agnostic.

Index Terms—adaptive assessment, dynamic rating, Elo, Glicko, TrueSkill, knowledge tracing, graph-based recommendation, remediation, performance prediction, educational data mining

I. INTRODUCTION AND MOTIVATION

Standard Elo [?] updates a learner’s rating R after each item by $\Delta R = K(S - E)$, where K is a fixed constant, $S \in \{0, 1\}$ is the binary outcome, and E is the expected probability of success. This formulation has three critical deficiencies in an adaptive learning context.

Fixed uncertainty. A new learner’s rating is highly uncertain; a learner with 500 answered items is much better calibrated. Fixed K treats both identically, causing noisy oscillation for beginners and sluggish adaptation for advanced learners. Glicko [?] and Glicko-2 [?] address this with a rating deviation σ , but neither incorporates domain-specific signals.

Binary outcome signal. Pass/fail discards rich information: did the learner answer in 5 seconds or 5 minutes? Did they use a hint? Were they on a 6-item correct streak? Valorant’s ranked system uses individual performance rating (IPC score)

beyond just win/loss to modulate rank rate [?]; we apply the same principle to items.

Static difficulty. Fixed difficulty labels (K_d dependent only on Easy/Medium/Hard) ignore the structural importance of an item in the knowledge graph. An item at a high-centrality hub is pedagogically more diagnostic than a peripheral item of the same labeled difficulty.

DARS resolves all three deficiencies through a principled synthesis of psychometric rating theory, competitive matchmaking mechanics, and knowledge graph structure. The contributions of this paper are:

- C1. Dynamic uncertainty-decaying K :** a closed-form update factor that decays with item count and rating variance, collapsing to a stable floor for settled learners.
- C2. Multi-signal performance quality score $\mathcal{P}$:** a continuous $[0, 1]$ signal replacing binary S , encoding correctness, time efficiency, hint penalty, and streak state.
- C3. Streak multiplier ϕ :** a bounded multiplicative bonus/penalty applied to ΔR based on sustained performance trends.
- C4. Graph-centrality difficulty calibration:** item betweenness centrality modulates update magnitude, making hub-item responses more diagnostic.
- C5. Full system integration:** the rating model drives a knowledge graph router, remediation engine, velocity detector, and performance predictor — all formally specified and domain-agnostic.

II. DYNAMIC ADAPTIVE RATING SYSTEM (DARS)

A. Overview

Fig. 1 illustrates the DARS rating pipeline. For each submitted response, five signals feed the rating update: the binary outcome S , the time efficiency ratio τ , the hint usage indicator H , the streak state ϕ , and the item’s graph centrality $c(q)$. These combine into a performance quality score $\mathcal{P}$ and a graph-calibrated effective difficulty, which drive the dynamic K -factor update.

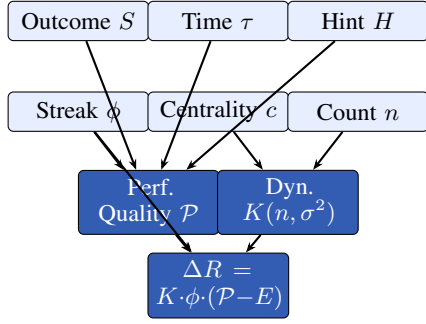


Fig. 1. DARS rating update pipeline. Five learner and item signals combine into performance quality $\mathcal{P}$, dynamic factor K , and streak multiplier ϕ to produce ΔR .

B. Performance Quality Score $\mathcal{P}$

The binary outcome $S \in \{0, 1\}$ is replaced by a continuous performance quality score $\mathcal{P} \in [0, 1]$ that encodes *how well* the learner answered, not just *whether* they were correct.

Definition 1 (Performance Quality Score).

$$\mathcal{P}(q) = \underbrace{S}_{\text{correctness}} \cdot \left(\underbrace{w_\tau \cdot \tau(q)}_{\text{time efficiency}} + \underbrace{w_H \cdot (1 - H(q))}_{\text{hint penalty}} + \underbrace{w_s \cdot \psi(q)}_{\text{streak quality}} \right) + (1 - S) \cdot \varepsilon_{\text{fail}} \quad (1)$$

where weights satisfy $w_\tau + w_H + w_s = 1$ (defaults: $w_\tau = 0.50$, $w_H = 0.30$, $w_s = 0.20$), and the components are defined as follows.

a) *Time efficiency* $\tau(q)$:

$$\tau(q) = \text{clip}\left(\frac{t_{\text{ref}}(q) - t_{\text{spent}}(q) + t_{\text{slack}}}{t_{\text{ref}}(q)}, 0, 1\right) \quad (2)$$

where $t_{\text{ref}}(q)$ is the reference solve time for item q (estimated from stem length, option count, and complexity; see Section V), and $t_{\text{slack}} = 0.15 \cdot t_{\text{ref}}$ allows a 15% tolerance before penalising. $\tau = 1$ for a very fast correct answer; $\tau \rightarrow 0$ as $t_{\text{spent}} \gg t_{\text{ref}}$.

b) *Hint penalty* $H(q)$:

$$H(q) = \min\left(1, \frac{n_{\text{hints}}(q)}{n_{\text{hints,max}}}\right) \in [0, 1] \quad (3)$$

where $n_{\text{hints}}(q)$ is the number of hints used on item q and $n_{\text{hints,max}}$ is the maximum available. A correct answer with no hints has $H = 0$ (no penalty); full hint usage yields $H = 1$ (maximum penalty).

c) *Streak quality* $\psi(q)$:

$$\psi(q) = \frac{1}{2} \left(1 + \frac{k}{\max(|k|, k_{\text{max}})} \right) \in [0, 1] \quad (4)$$

where k is the current signed streak ($k > 0$: consecutive correct; $k < 0$: consecutive incorrect) and $k_{\text{max}} = 5$ is the saturation streak length. A +5 streak yields $\psi = 1.0$; neutral ($k = 0$) yields $\psi = 0.5$; a -5 streak yields $\psi = 0.0$.

d) *Failure floor* $\varepsilon_{\text{fail}}$: On an incorrect response ($S = 0$), $\mathcal{P}$ is set to a small positive value rather than zero to preserve a residual signal:

$$\varepsilon_{\text{fail}} = \text{clip}(0.05 \cdot \tau(q), 0, 0.10) \quad (5)$$

This rewards a learner who attempted a hard item slowly (good faith) marginally more than one who guessed instantly on an easy item.

C. Dynamic Update Factor $K(n, \sigma^2)$

Standard Elo's fixed K creates two failure modes: oversensitivity for new learners (too much noise) and undersensitivity for calibrated learners (too slow to adapt). DARS uses a *three-phase dynamic* K inspired by Glicko-2's rating deviation and competitive gaming's provisional systems.

Definition 2 (Dynamic Update Factor).

$$K(n, \sigma^2) = K_{\min} + (K_{\max} - K_{\min}) \cdot \underbrace{e^{-\lambda_n \cdot \max(0, n - n_{\text{prov}})}}_{\text{count decay}} \cdot \underbrace{\left(\frac{\sigma^2}{\sigma^2 + \sigma_0^2} \right)}_{\text{uncertainty weight}} \quad (6)$$

where:

- n = total items answered by this learner to date.
- $n_{\text{prov}} = 30$: provisional phase length (first 30 items, K stays at $K_{\max}$).
- $K_{\max} = 60$: maximum K during provisional phase (high sensitivity for fast early calibration).
- $K_{\min} = 16$: floor K for fully calibrated learners (prevents stagnation).
- $\lambda_n = 0.015$: decay rate post-provisional.
- σ^2 : empirical variance of ΔR over the last 20 items (rolling window), measuring rating volatility.
- $\sigma_0^2 = 100$: reference variance (half-saturation constant); if $\sigma^2 \gg \sigma_0^2$, the learner is volatile and K stays high.

Proposition 1 (Convergence of K). As $n \rightarrow \infty$ and $\sigma^2 \rightarrow 0$ (learner stabilises at true ability), $K(n, \sigma^2) \rightarrow K_{\min}$. Conversely, a sudden performance shift (large σ^2) prevents K from collapsing, allowing re-calibration.

The three operational phases are:

$$\text{Phase}(n) = \begin{cases} \text{Provisional} & n < n_{\text{prov}} \quad [K \approx K_{\max}] \\ \text{Calibration} & n_{\text{prov}} \leq n < n_{\text{cal}} \quad [K \text{ decaying}] \\ \text{Established} & n \geq n_{\text{cal}} \quad [K \approx K_{\min}] \end{cases} \quad (7)$$

with $n_{\text{cal}} = 150$ (approximately one month of regular practice).

D. Streak Multiplier ϕ

Analogous to Valorant's win-streak RR bonus and loss-streak demotion shield, DARS applies a bounded streak multiplier to

ΔR :

$$\phi(k) = \begin{cases} 1 + \phi_{\max} \cdot \frac{k - k_{\text{thr}}^+}{k_{\max} - k_{\text{thr}}^+} & k \geq k_{\text{thr}}^+ \quad (\text{hot streak bonus}) \\ 1 & |k| < k_{\text{thr}} \quad (\text{neutral}) \\ 1 - \phi_{\max} \cdot \frac{|k| - k_{\text{thr}}^-}{k_{\max} - k_{\text{thr}}^-} & k \leq -k_{\text{thr}}^- \quad (\text{cold streak penalty}) \end{cases} \quad (8)$$

with parameters $\phi_{\max} = 0.35$, $k_{\text{thr}}^+ = k_{\text{thr}}^- = 3$, $k_{\max} = 7$. This yields $\phi \in [0.65, 1.35]$: a learner on a 7-item correct streak receives 35% amplified gains; one on a 7-item incorrect streak has penalties amplified by 35%.

Remark 1. ϕ is applied after K and $\mathcal{P}$ are computed, so it multiplicatively scales the already context-aware update. This mirrors how Valorant’s win-streak bonus amplifies, but does not replace, the base RR change.

E. Graph-Centrality Difficulty Calibration

Standard difficulty labels (Easy/Medium/Hard) are coarse. DARS refines the effective difficulty using the item’s structural importance in the knowledge graph.

Definition 3 (Effective Item Rating).

$$R_q^{\text{eff}} = R_q^{\text{label}} + \underbrace{\xi \cdot c_{\text{norm}}(q)}_{\text{centrality bonus}} \quad (9)$$

where R_q^{label} is the item’s base rating assigned by its difficulty label, $c_{\text{norm}}(q) \in [0, 1]$ is the normalised betweenness centrality of item q in the knowledge graph G , and $\xi = 150$ is the maximum centrality bonus (in rating points). A hub item at the $c_{\text{norm}} = 1.0$ percentile is treated as 150 rating points harder than its label alone suggests, because a correct answer on it is more diagnostic.

F. Full DARS Rating Update

Combining all components, the DARS update for item q is:

Definition 4 (DARS Update Rule).

$$E(q) = \frac{1}{1 + 10^{(R_q^{\text{eff}} - R^{(i-1)})/400}} \quad (10)$$

$$\Delta R_{\text{DARS}}(q) = K(n, \sigma^2) \cdot \phi(k) \cdot (\mathcal{P}(q) - E(q)) \quad (11)$$

$$R^{(i)} = R^{(i-1)} + \Delta R_{\text{DARS}}(q) \quad (12)$$

Key properties:

- When $\mathcal{P} > E$: the learner outperformed expectations $\Rightarrow$ rating increases.
- When $\mathcal{P} < E$: the learner underperformed $\Rightarrow$ rating decreases.
- When $\mathcal{P} = E$: no change — the item was exactly as expected.
- K and ϕ scale the magnitude without altering the direction.

G. Comparison with Baseline Rating Systems

Table I positions DARS against related systems.

TABLE I
FEATURE COMPARISON OF RATING SYSTEMS

Feature	Elo	Glicko-2	TrueSkill	Val. RR	DARS
Dynamic K	×	✓	✓	×	✓
Uncertainty model	×	✓	✓	×	✓
Provisional phase	×	✓	✓	✓	✓
Streak multiplier	×	×	×	✓	✓
Performance quality	×	×	×	✓	✓
Time signal	×	×	×	×	✓
Hint signal	×	×	×	×	✓
Graph calibration	×	×	×	×	✓
Domain-agnostic	✓	✓	✓	×	✓

TABLE II
DARS SITTING TRACE ($R^{(0)}=1500$, PROVISIONAL PHASE)

i	S	Diff	τ	H	$\mathcal{P}$	ϕ	ΔR	$R^{(i)}$
1	1	Easy	0.80	0	0.82	1.00	+5.2	1505.2
2	1	Medium	0.60	0	0.74	1.00	+9.6	1514.8
3	0	Easy	0.45	0	0.02	1.00	−14.7	1500.1
4	0	Hard	0.60	1	0.03	1.00	−4.8	1495.3
5	1	Easy	0.95	0	0.88	1.00	+11.3	1506.6
6	1	Easy	0.78	0	0.81	1.08	+11.7	1518.3
7	1	Medium	0.65	0	0.76	1.18	+16.2	1534.5
8	1	Medium	0.55	0	0.71	1.29	+14.8	1549.3
9	0	Easy	0.20	0	0.01	1.00	−21.4	1527.9
10	0	Medium	0.88	0	0.04	1.00	−17.3	1510.6

H. Dynamic Sitting Trace Example

Table II traces DARS across a 10-item sitting with $R^{(0)} = 1500$, $n = 0$ (provisional, $K \approx 60$), $k = 0$ initially.

Key observations:

- **Row 5 vs row 1:** both Easy correct, but row 5’s faster time ($\tau = 0.95$ vs 0.80) yields $\mathcal{P} = 0.88$ vs 0.82, earning more rating (+11.3 vs +5.2).
- **Row 6–8:** a 3-item correct streak activates $\phi > 1$ (1.08 $\rightarrow$ 1.29), amplifying gains on consecutive correct responses.
- **Row 9:** an easy item failed after the streak resets $\phi = 1.00$ and incurs a large penalty (−21.4) because $\mathcal{P} \approx 0$ (very fast incorrect on easy item).
- **Row 4 vs row 3:** failing hard (−4.8) costs far less than failing easy (−14.7), because E is high for hard items.

I. Rating Tiers

$$\text{tier}(R) = \begin{cases} \text{Bronze} & R < 1600 \\ \text{Silver} & 1600 \leq R < 2000 \\ \text{Gold} & 2000 \leq R < 2500 \\ \text{Platinum} & R \geq 2500 \end{cases} \quad (13)$$

III. KNOWLEDGE GRAPH AND ITEM ROUTING

A. Item Knowledge Graph

Definition 5 (Item Graph). $G = (V, E, w)$ is a directed weighted graph. Each node $v \in V$ represents an assessment item with attributes (domain_v , topic_v , R_v^{label} , $t_{\text{ref},v}$, c_v). Edge weight $w(u, v) \in [0, 1]$ encodes pedagogical similarity (lower = closer):

$$w(u, v) = \alpha_1 d_R + \alpha_2 d_{\text{marks}} + \alpha_3 d_{\text{topic}} + \alpha_4 d_{\text{type}}, \quad \sum \alpha_i = 1 \quad (14)$$

Three edge types encode the knowledge structure: **same-topic** (e_s), **domain-flow** (e_f , prerequisite order), and **domain-bridge** (e_b , cross-domain dependency).

B. Learner State

Definition 6 (Learner State).

$$\mathbf{s} = (R, n, \sigma^2, k, m, W, \mathcal{Q}_{\text{ans}}, \mathcal{Q}_{\text{srv}}, \pi_{\text{rem}}) \quad (15)$$

where R is current DARS rating; n is total items answered; σ^2 is rolling rating variance (last 20 items); k is current signed streak; $m \in \{\text{hot}, \text{steady}, \text{cold}\}$ is momentum; W is the weak-topic set (per-topic accuracy $< \tau_w$); $\mathcal{Q}_{\text{ans}}$, $\mathcal{Q}_{\text{srv}}$ exclude answered and served items; π_{rem} tracks remediation progress.

Note the addition of n , σ^2 , and k to the state compared to prior work: these are required by the dynamic K and ϕ computations.

C. Momentum

$$m = \begin{cases} \text{hot} & k \geq 5 \wedge \hat{a} \geq 0.80 \\ \text{cold} & k \leq -3 \wedge \hat{a} < 0.50 \\ \text{steady} & \text{otherwise} \end{cases} \quad (16)$$

D. Target Policy

$$R_{\text{tgt}} = R + \Delta(m), \quad \Delta = \begin{cases} +100 & m = \text{hot} \\ +50 & m = \text{steady}, \hat{a} \geq 0.70 \\ 0 & \text{otherwise} \\ -100 & m = \text{cold} \end{cases} \quad (17)$$

R_{tgt} maps to a target tier via Eq. (13).

E. Neighbourhood Selection

a) *Standard neighbourhood:*

$$\mathcal{N}(q, k_h) = \{q' \in V : d_G(q, q') \leq k_h, q' \notin \mathcal{Q}_{\text{ans}}, q' \notin \mathcal{Q}_{\text{srv}}\}, \quad (18)$$

b) *Remediation neighbourhood:*

$$\mathcal{N}_r(q_m, h) = \{q' \in V : d_G(q_m, q') \leq h, q' \notin \mathcal{Q}_{\text{ans}}, \text{diff}(q') < \text{diff}(q_m)\}, \quad h = 3 \quad (19)$$

Algorithm 1 DARS-Driven Adaptive Item Recommendation

Require: State $\mathbf{s}$, graph G , $k_h=2$, $h=3$

Ensure: Item q^* and explanation

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1: Compute  $m$  via Eq. (16)
2:  $R_{\text{tgt}} \leftarrow R + \Delta(m)$  ▷ Eq. (17)
3:  $\rho_{\text{prov}} \leftarrow \min(1, n/n_{\text{prov}})$ 
4: if  $\pi_{\text{rem}}.\text{active}$  then
5:    $\mathcal{C} \leftarrow \mathcal{N}_r(\pi_{\text{rem}}.q_m, h)$ 
6: else
7:    $\mathcal{C} \leftarrow \mathcal{N}(q_{\text{last}}, k_h)$ 
8: end if
9: if  $\mathcal{C} = \emptyset$  then
10:   $\mathcal{C} \leftarrow$  fallback: unanswered items nearest  $R_{\text{tgt}}$ 
11: end if
12: for each  $q' \in \mathcal{C}$  do
13:   Compute  $h(q')$ ,  $g(q')$ ,  $R_{q'}^{\text{eff}}$ 
14:    $\text{score}(q') \leftarrow w_h h(q') + w_g g(q') \cdot \rho_{\text{prov}}$ 
15: end for
16:  $q^* \leftarrow \arg \max_{q' \in \mathcal{C}} \text{score}(q')$ 
17: Attach explanation: hop distance, tier,  $R_{\text{tgt}}$ ,  $\phi$ , bonuses
18: return  $q^*$ 

```

F. DARS-Aware Candidate Scoring

The item scoring function is now DARS-aware: it uses the effective item rating $R_{q'}^{\text{eff}}$ (Eq. 9) for difficulty distance, and the learner's provisional status to down-weight peripheral items during early calibration.

$$\text{score}(q') = w_h \cdot h(q') + w_g \cdot g(q') \cdot \rho_{\text{prov}}(n) \quad (20)$$

where $\rho_{\text{prov}}(n) = \min(1, n/n_{\text{prov}})$ down-weights the graph-structural component during the provisional phase (favouring topic-local items when the learner is uncalibrated).

1) *Heuristic component:*

$$h(q') = 1 - \frac{w_1 d_{\text{diff}}(q') + w_2 d_R^{\text{eff}}(q')}{C_{\text{norm}}} + \beta_w \mathbb{I}[q' \in W] + \beta_t \mathbb{I}[\text{type}(q') \neq \text{type}(q)] \quad (21)$$

with $d_R^{\text{eff}}(q') = |R_{q'}^{\text{eff}} - R_{\text{tgt}}|$.

2) *Graph-structural boost:*

$$g(q') = \gamma_1 \rho_{\text{in}}(q') + \gamma_2 \rho_{\text{out}}(q') + \gamma_3 c_{\text{norm}}(q') \quad (22)$$

G. Full Item Recommendation Algorithm

IV. REMEDIATION AND RETRY ENGINE

An incorrect response flags item q_m as an *unresolved miss*. DARS applies a reduced K during remediation ($K_{\text{rem}} = 0.6 \cdot K$) to prevent the easier remediation items from over-updating the rating downward.

Exit criterion:

$$n_{\text{rem}} \geq N_r \wedge \hat{a}_r \geq A_r \quad (N_r = 3, A_r = 0.75) \quad (23)$$

Algorithm 2 DARS Remediation and Retry

Require: Missed item q_m , $N_r=3$, $A_r=0.75$

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1: Flag  $q_m$  UNRESOLVEDMISS;  $\pi_{\text{rem}} \leftarrow \text{active}$ 
2:  $n_{\text{rem}} \leftarrow 0$ ;  $\hat{a}_r \leftarrow 0$ ; save  $K_{\text{saved}} \leftarrow K(n, \sigma^2)$ 
3:  $K \leftarrow 0.6 \cdot K_{\text{saved}}$   $\triangleright$  attenuate during remediation
4: repeat
5:    $q' \leftarrow \text{Alg. 1 with } \mathcal{C} = \mathcal{N}_r(q_m, h)$ 
6:   Serve  $q'$ ; update  $n_{\text{rem}}$ ,  $\hat{a}_r$ ,  $R$  via Eq. (11)
7: until  $n_{\text{rem}} \geq N_r \wedge \hat{a}_r \geq A_r$ 
8:  $K \leftarrow K_{\text{saved}}$ ; queue  $q_m$  for retry
9: if learner answers  $q_m$  correctly then
10:   Clear UNRESOLVEDMISS; apply full  $\Delta R$  with  $\phi$  reset
11: else
12:   Re-activate  $\pi_{\text{rem}}$ ; carry miss to next session
13: end if
```

V. ANSWER-VELOCITY ANOMALY DETECTION

A. Reference Time Estimation

$$t_{\text{ref}}(q) = \alpha_0 + \alpha_1 \ell_{\text{stem}}(q) + \alpha_2 \ell_{\text{opts}}(q) + \alpha_3 c_{\text{idx}}(q) \quad (24)$$

with $\alpha_0 = 10$ s, $\alpha_1 = 0.02$ s/char, $\alpha_2 = 0.01$ s/char, $\alpha_3 = 30$ s for high-complexity items.

B. Anomaly Condition and Action

The time efficiency $\tau(q)$ (Eq. 2) already enters $\mathcal{P}$, so fast correct answers are naturally penalised. An *anomalous* response is one where $t_{\text{spent}} < t_{\text{thresh}}(q)$ with:

$$t_{\text{thresh}}(q) = 0.25 \cdot t_{\text{ref}}(q) \quad (25)$$

i.e. the learner used less than 25% of the reference time. In this case, even $\mathcal{P}$ may be inflated by the fast time; an additional hard cap is applied:

$$\Delta R_{\text{DARS}}^{\text{capped}}(q) = \min\left(\Delta R_{\text{DARS}}(q), K_{\min} \cdot 0.10\right) \quad \text{if } t_{\text{spent}} < t_{\text{thresh}} \quad (26)$$

capping the maximum gain at $0.10 \cdot K_{\min} = 1.6$ rating points for anomalous responses, regardless of $\mathcal{P}$.

VI. LEARNER RISK AND INTERVENTION

A. Composite Risk Score

$$\text{Risk}(s) = w_1(1 - \text{acc}(s)) + w_2(1 - \text{comp}(s)) + w_3 \mathbb{I}[R(s) < R_{\text{floor}}] + w_4 \mathbb{I}[\sigma^2(s) > \sigma_{\text{volatile}}^2] \quad (27)$$

with $w_1 = 0.45$, $w_2 = 0.25$, $w_3 = 0.20$, $w_4 = 0.10$, and $\sigma_{\text{volatile}}^2 = 400$. The fourth term is new: a learner whose rating is highly volatile (inconsistent performance) is flagged even if their mean accuracy appears acceptable.

B. Weakness Severity

$$\text{sev}(t_i) = (1 - \text{acc}_s(t_i)) \cdot w_{\text{topic}}(t_i) \quad (28)$$

Algorithm 3 Intervention Recommendation

Require: Cohort $\mathcal{S}$, volatility threshold σ_v^2

```

1: for each  $s \in \mathcal{S}$  do
2:   Compute Risk( $s$ ) via Eq. (27)
3: end for
4:  $\mathcal{S}_r \leftarrow \{s : \text{Risk}(s) \geq 0.70\}$ 
5: for each  $s \in \mathcal{S}_r$  do
6:    $t^* \leftarrow \arg \max_{t_i \in W(s)} \text{sev}(t_i)$ 
7:   if  $\sigma^2(s) > \sigma_v^2$  then
8:     Flag as VOLATILE; recommend consistency drills
       before topic expansion
9:   else
10:    Generate targeted practice on  $t^*$ 
11:   end if
12:   Schedule follow-up in  $\Delta t_{\text{followup}}$  days
13: end for
```

C. Intervention Algorithm

VII. PERFORMANCE PREDICTION MODEL

A. Readiness Gate

$$\text{READY} \Leftrightarrow R \geq R_{\text{pred_min}} \wedge n \geq n_{\text{cal}} \wedge \bigwedge_j \text{comp}_j \geq \theta \quad (29)$$

Adding $n \geq n_{\text{cal}}$ ensures the learner has exited the provisional phase before any prediction is surfaced.

B. Five-Component Basis

We extend the prior four-component basis with a fifth signal: **rating stability** V_n , directly available from DARS.

$$B = w_1 A_n + w_2 P_n + w_3 C_n + w_4 I_n + w_5 V_n \in [0, 1] \quad (30)$$

C1 – Accuracy (A_n):

$$A_n = \min\left(1, \frac{a_{\text{raw}} - a_f}{a_c - a_f}\right) \quad (31)$$

C2 – Proficiency (P_n):

$$P_n = \frac{R - R_b}{R_m - R_b} \quad (32)$$

C3 – Domain Consistency (C_n):

$$C_n = \max\left(0, 1 - \frac{\sigma_{\text{topic}}}{\bar{a} + \varepsilon}\right) \quad (33)$$

C4 – Improvement Trend (I_n):

$$I_n = \frac{\Delta a + \delta}{2\delta}, \quad \Delta a \in [-\delta, +\delta] \text{ (clipped)} \quad (34)$$

C5 – Rating Stability (V_n):

$$V_n = \max\left(0, 1 - \frac{\sigma^2}{\sigma_{\text{ref}}^2}\right) \quad (35)$$

where $\sigma_{\text{ref}}^2 = 400$ is the reference volatility above which a learner is considered unstable. A learner with low rating

TABLE III
FIVE-COMPONENT PERFORMANCE PREDICTION BASIS

	Component	Normalised Formula	Range	w_i
C1	Accuracy	$\min(1, (a - a_f)/(a_c - a_f))$	$[0, 1]$	0.35
C2	Proficiency	$(R - R_b)/(R_m - R_b)$	$[0, 1]$	0.25
C3	Consistency	$\max(0, 1 - \sigma_t/(\bar{a} + \varepsilon))$	$[0, 1]$	0.20
C4	Improvement	$(\Delta a + \delta)/(2\delta)$	$[0, 1]$	0.10
C5	Stability	$\max(0, 1 - \sigma^2/\sigma_r^2)$	$[0, 1]$	0.10

variance ($\sigma^2 \ll \sigma_{\text{ref}}^2$) scores $V_n \approx 1$; a volatile learner scores $V_n \approx 0$, reducing the confidence in the prediction. Default weights: $(w_1, w_2, w_3, w_4, w_5) = (0.35, 0.25, 0.20, 0.10, 0.10)$.

Proposition 2. *Since $w_1 + \dots + w_5 = 1$ and each $C_i \in [0, 1]$, we have $B \in [0, 1]$ for all valid inputs.*

Table III summarises all five components.

C. Score Band and Confidence

$$\hat{S} = S_{\min} + (S_{\max} - S_{\min}) \cdot B \quad (36)$$

$$S_{\text{lo}} = \hat{S} - \eta(1 - B)(1 + \nu(1 - V_n)) \quad (37)$$

$$S_{\text{hi}} = \hat{S} + \eta B \cdot V_n \quad (38)$$

where $\eta = 10$ is the base band half-width and $\nu = 0.5$ amplifies the lower bound uncertainty for volatile learners. A stable high- B learner gets a tight, symmetric band; a volatile low- B learner gets a wider, asymmetric band.

$$\Gamma = 0.40 B + 0.30 V_n + 0.20 \text{CompRate} + 0.10 \min(1, n/n_{\text{cal}}) \quad (39)$$

The fourth term rewards learners who have exited the provisional phase, grounding confidence in well-calibrated ratings.

D. Worked Example

Input: $a = 0.78$, $R = 2350$, $n = 200$, $\sigma^2 = 85$, topic accuracies $[0.85, 0.72, 0.70, 0.75, 0.80]$ ($\bar{a} = 0.764$, $\sigma_t = 0.060$), $\Delta a = +0.07$, $\text{CompRate} = 0.70$. Scale $[0, 100]$; $a_f = 0.50$, $a_c = 0.85$, $R_b = 1600$, $R_m = 2800$, $\delta = 0.15$, $\sigma_r^2 = 400$.

$$A_n = \min(1, 0.28/0.35) = 0.800$$

$$P_n = 750/1200 = 0.625$$

$$C_n = 1 - 0.060/0.774 = 0.922$$

$$I_n = 0.22/0.30 = 0.733$$

$$V_n = \max(0, 1 - 85/400) = 0.788$$

$$\begin{aligned} B &= 0.35(0.800) + 0.25(0.625) + 0.20(0.922) + 0.10(0.733) + 0.10(0.788) \\ &= 0.280 + 0.156 + 0.184 + 0.073 + 0.079 = \mathbf{0.772} \end{aligned}$$

$$\hat{S} = 100 \cdot 0.772 = 77.2$$

$$S_{\text{lo}} = 77.2 - 10(0.228)(1 + 0.5 \cdot 0.212) = 74.8$$

$$S_{\text{hi}} = 77.2 + 10 \cdot 0.772 \cdot 0.788 = 83.3$$

$$\Gamma = 0.40(0.772) + 0.30(0.788) + 0.20(0.70) + 0.10 \cdot 1.0 = 0.883$$

Result: predicted score 75–83 (best estimate 77.2/100), confidence 88.3%.

VIII. COACHING ENGINE

A. Proficiency and Volatility Classification

$$\text{prof} = \begin{cases} \text{Low} & R < R_{\text{low}} \\ \text{Medium} & R_{\text{low}} \leq R \leq R_{\text{high}} \\ \text{High} & R > R_{\text{high}} \end{cases} \quad \text{vol} = \begin{cases} \text{Stable} & \sigma^2 < 100 \\ \text{Moderate} & 100 \leq \sigma^2 \\ \text{Volatile} & \sigma^2 \geq 300 \end{cases} \quad (40)$$

B. Recommendation Rules

$$\text{advice} = \begin{cases} \text{Consolidate fundamentals} & \text{prof} = \text{Low}, a < 0.50 \\ \text{Consistency drills, same tier} & \text{vol} = \text{Volatile} \\ \text{Progress to harder items} & \text{prof} = \text{Med, improving} \\ \text{Speed + timed sets} & \text{prof} = \text{High}, a > 0.75 \\ \text{Review topics, reduce difficulty} & \text{declining trend} \end{cases} \quad (41)$$

The volatility rule is new: a volatile learner is steered towards consistency-building exercises at their current tier rather than difficulty progression, regardless of proficiency level.

IX. PARAMETER REFERENCE

TABLE IV
COMPLETE DARS PARAMETER REFERENCE

Module	Parameter	Default	Description
DARS Rating	$K_{\max}$	60	Max update factor (provisional)
	$K_{\min}$	16	Floor update factor (established)
	n_{prov}	30	Provisional phase length
	n_{cal}	150	Calibration complete threshold
	λ_n	0.015	Post-provisional decay rate
	σ_0^2	100	Volatility half-saturation
	ξ	150	Max centrality rating bonus
Perf. Quality	w_τ	0.50	Time efficiency weight
	w_H	0.30	Hint penalty weight
	w_s	0.20	Streak quality weight
	$\varepsilon_{\text{fail}}$	≤ 0.10	Failure floor
Streak	$\phi_{\max}$	0.35	Max streak multiplier
	k_{thr}	3	Streak activation threshold
	$k_{\max}$	7	Streak saturation length
Graph Rec.	k_h	2	Std. hop limit
	h	3	Remed. hop limit
	τ_w	0.60	Weak-topic threshold
Remediation	N_r	3	Min. steps before retry
	A_r	0.75	Min. accuracy before retry
Velocity	t_{thresh}	25%	Anomaly time fraction
	σ_v^2	400	Volatility intervention flag
Predictor	$(w_1, \dots, w_5)$	see §VII	Component weights
	θ	0.70	Completion gate
	η	10	Band half-width
	ν	0.50	Volatile band amplifier